Testora

Aptitude Preparation & Online Test System — Project Overview for Students, Teachers and Admins

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Testora is a web platform for aptitude test preparation used by three kinds of people — **students** who prepare and take tests, **teachers** who create and assign quizzes to their students, and **admins** who manage the platform's content and users. It runs as a React web app talking to a Node.js/Express API backed by MongoDB.

How content is organized

Everything test-related sits under a simple three-level structure: **Category** (subject) → **Topic** → **Question**. For example: Quantitative Aptitude → Percentages → “What is 20% of 150?”. Admins create Categories and Topics. Questions are then attached to a Topic (and therefore to a Category). Teachers don't manage this structure directly — they just pick a Category and a Topic when writing quiz questions.

Two ways students get tested

	Mock Test	Quiz
Created by	Admin	Teacher
Who can take it	Any student, once published	Only students it was assigned to
Question source	Picked from the existing question bank	Written fresh, inline, during quiz creation
Visibility	Shown to all students once published	Shown only within its start–end date window

Both are scored the same way: **marks per question**, with an **optional negative mark** deducted for each wrong answer (skipped questions score zero, never negative). A student's result always shows marks obtained out of the maximum possible, not just “number correct.”

Who can do what (role-based access)

- A logged-in session is a JWT token tied to one role — student, teacher, or admin — fixed at account creation and never changeable by the user.
- Every API endpoint checks both “are you logged in” and “is your role allowed here.” Students, for example, can never fetch a question's correct answer through the API before they submit an attempt.
- If an admin deactivates an account, that account is locked out of every protected action immediately — even if already logged in with a valid token — and the next action in the browser sends them back to the login screen.

Student Portal

*Everything under **/student/...**. A student's job here is: learn a topic, practice it, take tests, and track progress.*

Screen	What it's for
Dashboard	Landing page after login. Shows a stats snapshot (tests taken, questions solved, average accuracy), the student's weakest topics with a one-click link to practice them, any quizzes currently assigned and available, and the most recent test results.
Topics	Browse subjects (Quantitative Aptitude, Logical Reasoning, Verbal Ability, plus anything else admins have added) and drill into a subject to see its topics.
Topic Detail	One topic's page: description, how many study notes and how many questions exist for it, the study notes themselves (formulas, worked examples, key points), and a button to start practicing that topic.
Practice	Untimed, one-question-at-a-time practice for a chosen topic. Submit an answer and get instant feedback — correct/incorrect, the right answer, and an explanation — before moving to the next question. Ends with a session summary (correct/wrong/skipped, accuracy).
Mock Tests	Lists every mock test an admin has published, with question count and duration. Already-attempted tests show “View Result” instead of “Start,” since each mock test can only be attempted once.
Mock Test Exam	The actual timed test screen — countdown timer, one question at a time with a navigator, answers saved as you go, auto-submits when time runs out. Correct answers are never sent to the browser until after you submit.
My Quizzes	Quizzes a teacher has assigned, each tagged Upcoming / Available / Completed / Expired based on the quiz's scheduled date window. Only “Available” ones can be started.
Quiz Exam	Same timed exam experience as a Mock Test, just for a teacher-assigned quiz — scored using that quiz's configured marks and negative marking.
Result Page	After submitting any test: accuracy %, correct/wrong/skipped counts, time taken, marks obtained out of maximum possible marks, a per-topic accuracy breakdown, and a full question-by-question review (your answer vs. the correct one, with explanation) — only ever shown after submission.
Performance	The “how am I doing overall” page — accuracy trends across all tests taken, a chart of accuracy per topic, and topics automatically flagged as Weak / Needs Practice / Strong based on past attempts, each with a direct link into Practice mode.
Test History	A simple, filterable, paginated log of every test ever attempted (mock test or quiz), with a link back to each one's full result.
Profile	Edit name, mobile, college, course, and year; change password.

Teacher Portal

*Everything under **/teacher/...**. A teacher's job here is: write quizzes and see how their students did.*

Screen	What it's for
Dashboard	Landing page — quick stats (total quizzes, how many are published vs. still drafts) and a shortcut list of recent quizzes with a link straight to each one's results.
Quiz Management	Every quiz this teacher has created: question count, total marks (and negative-marking amount if enabled), duration, draft/published status, and the date window it's open for. Actions: view results, or delete (with a confirmation prompt).
Create Quiz	A guided 3-step quiz builder: **1. Select Category** — pick the subject this quiz belongs to. **2. Quiz Details** — title, description, how many questions, duration, marks per question, and an optional negative-marking amount, plus the start/end date window. **3. Set Questions** — one form per question (as many as chosen in step 2), each needing a topic, the question text, four options, which one is correct, and an optional explanation. Submitting creates all those questions fresh in the question bank, creates the quiz, and automatically assigns it to **every active student** — there's no manual student-picking step.
Quiz Results	For one quiz: how many students it was assigned to, how many have completed it, the class average accuracy, and a table of every student's score (marks obtained / max marks), accuracy, and completion status.
Profile	Edit name and area of specialization; change password.

Admin Portal

*Everything under **/admin/...**. An admin's job here is: manage who's on the platform and what content exists.*

Screen	What it's for
Dashboard	Platform-wide numbers pulled live from the database: Total Students, Total Teachers, Total Topics, Total Questions (plus mock test and test-attempt counts), and a feed of the most recent completed test attempts across everyone.
Students	Every registered student — name, college, course, year, and status — with an Activate/Deactivate action. Deactivating a student immediately locks them out of the app, including any session they're already logged into.
Teachers	Every teacher account — name, mobile, specialization, and status — with Edit and Activate/Deactivate actions. This is also where new teacher accounts get created (there's no public “sign up as a teacher” page — only an admin can create one), by setting their name, email, mobile, specialization, and an initial password.
Categories	Create/edit/deactivate the top-level subjects (e.g. “Quantitative Aptitude”). These are exactly the categories teachers see when creating a quiz.
Topics	Create/edit/deactivate topics, each one filed under a category (e.g. “Percentages” under Quantitative Aptitude).
Concepts	Study-note content attached to a topic — free-text explanation, a list of formulas, worked examples, and key points. This is separate from questions; it's what students read on a Topic Detail page before practicing.
Questions	The master question bank: full create/edit/deactivate for every question (text, category, topic, four options, correct answer, optional explanation), searchable and filterable by category and topic. Also where bulk content management happens: CSV import from a spreadsheet; JSON import using a documented format (pasted or uploaded as a .json file) that validates every question before importing anything and refuses the whole batch if even one is invalid or a duplicate; and CSV export, either everything or a filtered subset.
Mock Tests	Create/edit full-length mock tests: title, description, duration, marks per question, optional negative marking, and a question picker (filterable by category) to assemble the test. A publish/unpublish toggle controls whether students can see and attempt it.

How the pieces connect — a worked example

1. **Admin** creates the category “Quantitative Aptitude” and the topic “Percentages” under it.
2. **Admin** either writes individual questions on “Percentages” in the Question Bank, or bulk-imports 50 of them via CSV/JSON.
3. **Admin** picks 20 of those questions to assemble a published “Quantitative Aptitude Mock Test,” with negative marking enabled.
4. Any **student** can now see and attempt that mock test, once, under its timer — never seeing correct answers until after submitting — and gets a marks-based result (accounting for the negative marking) on their Result page and in their history.
5. Separately, a **teacher** runs their own class quiz: picks “Quantitative Aptitude” as the category, sets 10 questions, writes each one fresh (text, options, correct answer) scoped to topics like “Percentages” or “Profit & Loss,” and publishes it — which auto-assigns it to every active student.
6. The **teacher** later opens Quiz Results to see each student's score and class average; the **admin** sees that same activity reflected in the platform-wide dashboard counts and recent-attempts feed.

This document reflects the application as of this writing. For exact API routes, environment setup, and a running log of recent backend/frontend changes, see the main README.md in the project root.